

Review

Evidence-Based Dampness-Heat ZHENG (Syndrome) in Cancer: Current Progress toward Establishing Relevant Animal Model with Pancreatic Tumor*

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ABSTRACT Cancer is one of the deadliest diseases affecting the health of human beings. With limited therapeutic options available, complementary and alternative medicine has been widely adopted in cancer management and is increasingly becoming accepted by both patients and healthcare workers alike. Chinese medicine characterized by its unique diagnostic and treatment system is the most widely applied complementary and alternative medicine. It emphasizes symptoms and ZHENG (syndrome)-based treatment combined with contemporary disease diagnosis and further stratifies patients into individualized medicine subgroups. As a representative cancer with the highest degree of malignancy, pancreatic cancer is traditionally classified into the "amassment and accumulation". Emerging perspectives define the core pathogenesis of pancreatic cancer as "dampness-heat" and the respective treatment "clearing heat and resolving dampness" has been demonstrated to prolong survival in pancreatic cancer patients, as has been observed in many other cancers. This clinical advantage encourages an exploration of the essence of dampness-heat ZHENG (DHZ) in cancer and investigation into underlying mechanisms of action of herbal formulations against dampness-heat. However, at present, there is a lack of understanding of the molecular characteristics of DHZ in cancer and no standardized and widely accepted animal model to study this core syndrome *in vivo*. The shortage of animal models limits the ability to uncover the antitumor mechanisms of herbal medicines and to assess the safety profile of the natural products derived from them. This review summarizes the current research on DHZ in cancer in terms of the clinical aspects, molecular landscape, and animal models. This study aims to provide comprehensive insight that can be used for the establishment of a future standardized ZHENG-based cancer animal model.

KEYWORDS cancer, Chinese medicine syndrome, ZHENG, dampness-heat syndrome, animal model, pancreatic tumor, evaluation methods

Cancer is an intractable disease characterized by the uncontrolled growth and spread of abnormal cells. The American Cancer Society has conservatively estimated that 1,898,160 new cancer cases and 608,570 cancer deaths will occur in 2021 in the United States.⁽¹⁾ The morbidity and mortality of several highly malignant cancers are even more discouraging. For example, due to the asymptomatic nature of the pathogenesis of pancreatic cancer (PC), up to 53% of patients have already progressed to an advanced stage at the initial diagnosis and thus miss the opportunity for surgical resection. Its high malignancy leads to a 5-year relative survival rate of 9% for all stages combined and only 3% for patients at a distant stage.⁽²⁾ For this type of cancer, treatment resistance and a high recurrence rate are the main reasons for the shortened patients' survival time. Currently, surgery serves as the only potential curative strategy, and most patients are administered chemotherapy and

radiation as alleviative treatments. In addition, emerging targeted agents and immunotherapy remain to be tested. Despite the popularization of health education and the improvement in early detection and treatment of cancer, the low survival rate and poor quality of life faced by cancer patients remain significant threats to public health.

ZHENG (syndrome)-based classification,

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effectiveness assessments, and drug discovery have recently been explored as promising modalities for cancer management.^(3,4) Clinical studies have suggested that individualized adjuvant treatment based on *ZHENG* differentiation in Chinese medicine (CM) brought survival benefits to PC patients.^(5,6) In particular, the identification of the core pathogenesis of dampness-heat syndrome (DHZ) could contribute greatly to PC management.^(7,8) In the past two decades, studies on *ZHENG*, especially DHZ, have attracted increasing attention, from clinical observational studies, clinical trials, and molecular landscapes to animal models.⁽⁹⁻¹²⁾ Here, we extracted published papers from Chinese and English databases (China National Knowledge Infrastructure, SinoMed, Wanfang, VIP, PubMed, SpringerLink and ScienceDirect, from 1980 to 2022) using a combination of the words "cancer, dampness-heat (or Shi-re), clearing-heat (or Qing-re), resolving dampness (or Hua-shi), molecular mechanisms, and animal models" as search terms with the aims of comprehensively reviewing the research progress of DHZ in cancer.

Based on an analysis of the available literature, the present article attempted to explore a standard paradigm for establishing a DHZ model of cancer and model evaluation methods. We expect to delineate a reasonable *ZHENG* cancer model, and offer a future fundamental tool to investigate the essence of *ZHENG* and pharmacological mechanisms of active ingredients in herbal medicines prescribed according to the *ZHENG* in cancer. In other words, research on *ZHENG* or syndrome-guided medication must be conducted on a well-recognized model. Here, we provide comprehensive insight into the standardization of a DHZ model for cancer, to promote *ZHENG* studies and expedite CM modernization.

Clinical Advancement of DHZ in Cancer

The earliest application of CM *ZHENG* originated from the ancient experience of CM practitioners, and DHZ has always been valued in febrile illnesses. Updated CM practices emphasize the importance of dampness-heat in various types of cancer. As reported, Prof. LIU Lu-ming and his team from Fudan University Shanghai Cancer Center summarized their decades of practical experience and clinical research and proposed that "dampness and heat amassment" is the core pathogenesis of PC. Liu and his team regarded that internal dampness emerges from external dampness, and the resulting dampness obstructed in the body for an extended period can

generate heat. Thus, the intermingling of dampness and heat cause the development of DHZ. Accordingly, the therapeutic principle of "clearing heat and resolving dampness" was established. Qingyi Huaji Formula (清胰化积方) represents a successful prescription to treat PC with DHZ.⁽⁶⁾

A clinical observational study involving 28 PC patients revealed a statistically significant shortened overall survival (OS) among PC patients with the dampness-heat plus Pi (Spleen) deficiency syndrome (8 months, $n=9$) compared with those with DHZ alone (median OS of 18 months, $n=10$) and blood stasis syndrome (12 months, $n=10$, $P<0.05$).⁽¹³⁾ The other survival analysis of 107 PC cases suggested that patients treated with heat-clearing, diuresis promotion, and detoxification had a longer survival time than those treated with blood activating and stasis dissolving therapies or by tonifying qi and yang.⁽¹⁴⁾ Another retrospective study, including 232 cases, indicated that complementary treatment with the modified Qingyi Huaji Formula for more than 3 months could prolong the postoperative survival time of PC patients.⁽¹⁵⁾ In addition, 20 long-term surviving PC patients benefited from integrative treatment with CM and contemporary medicine, especially treatment for clearing heat and resolving dampness.⁽¹⁶⁾ These trials corroborate the validity of this therapeutic principle and formula against PC.

DHZ is also one of the main types of syndromes in individuals with liver cancer.⁽¹⁷⁾ Learning from the regularity of clinical treatment and inspired by the notion of pathogenesis from an insidious DHZ to the development of cancer, the principle of heat clearing, dampness resolving and detoxification was considered for individuals with liver cancer.^(11,18) In view of the importance of DHZ in liver cancer, the objective tongue inspection system has been developed to evaluate the effect of CM interventions.⁽¹²⁾ DHZ also plays a critical role in the tumorigenesis of colorectal cancer.⁽¹⁹⁾ In addition, clinical observation revealed a dynamic evolution from blood deficiency to both qi and yin deficiency and dampness and hot accumulation in patients with colorectal cancer.⁽²⁰⁾ Interestingly, the same DHZ with associated clinical indicators was found to exist both in patients with liver cancer and those with colorectal cancer.⁽²¹⁾ Moreover, clinical treatment based on DHZ for patients with lung cancer,⁽²²⁾ gastric cancer,⁽²³⁾ cancerous fever,⁽²⁴⁾ etc. is being extensively studied. Therefore, understanding the *ZHENG* nature of cancer and the modern connotation of syndrome-guided medication is essential for clinicians to best apply CM in clinical practice.

Molecular Landscape of DHZ in Cancer

Although considerable signs of research progress on proposed CM treatments according to clinical *ZHENG* differentiation have been noted in the field of cancer research, the molecular landscape associated with *ZHENG* remains unclear. As the most commonly identified *ZHENG* in PC, DHZ was reported to increase susceptibility to tumorigenesis and promote tumor progression by altering the inflammatory microenvironment, affecting gastrointestinal microecology, and inducing metabolic abnormalities.^(25,26) In 2010, a study showed that the presence of DHZ slowed the growth of PC, which was attributed to the low expression of chemokine receptor 5 (CCR5) and C-C motif chemokines 4 and 5 (CCL4 and CCL5) in PC tissue with DHZ.⁽¹⁰⁾ These results were challenged by a later research showing that different stages of tumor formation and the order of syndrome occurrence influenced tumor growth, and DHZ prior to tumor formation accelerated tumor growth.⁽²⁷⁾ A further study investigating alterations in the tumor microenvironment in PC patients with DHZ revealed an altered number of tumor-associated fibroblasts (CAFs) and tumor-associated macrophages (TAMs) infiltrating into the pancreatic tumor, which subsequently changed the levels of CAF- and TAM-related cytokines secretion, thereby affecting tumor growth and proliferation.⁽²⁸⁾

Despite controversy, existing studies at the very least have suggested that chemokines and the chemotactic aggregation of inflammatory cells were responsible for the DHZ in PC. Another study reported that the abundance of protumorigenic intestinal bacteria in PC patients with DHZ was higher than that in healthy volunteers,⁽²⁹⁾ and this class of protumor flora in the dampness-heat plus Pi deficiency syndrome group was even higher than that in the DHZ alone and blood stasis syndrome groups.⁽¹³⁾ The gut microenvironment under DHZ may contribute to PC progression, but the current research is inadequate and the precise mechanism remains to be elucidated. For colorectal cancer, one study suggested a significant difference in excision repair cross-complementing 1 (ERCC1) C19007T polymorphisms between patients with DHZ and those with Pi-Shen (Kidney) yang deficiency or Gan (Liver)-Shen yin deficiency.⁽³⁰⁾ In another study, patients with DHZ expressed higher mutant TP53, PCNA, PD-L1, Ki-67, CCL-2, interleukin (IL)-1 α , and cyclooxygenase 2 (COX2) than patients with Pi deficiency syndrome, which was linked with the poor prognosis of patients with DHZ.⁽³⁾

These comparative studies between different

syndromes highlighted that genetic variants may contribute more to DHZ than other syndromes in colorectal cancer. In addition, metabolic profiling analyses of postoperative colorectal cancer patients have revealed a potential correlation of altered creatinine and aminomalononic acid levels with the formation of DHZ.⁽³¹⁾ For mammary tumors induced by DMBA, dampness-heat induction promotes tumor growth by upregulating the expression of matrix metalloproteinase 9 (MMP9), tissue inhibitor of metalloproteinase-1 (TIMP-1), tumor necrosis factor α (TNF- α) and IL-1 β , implicating the involvement of metalloproteinases and inflammatory factors in breast cancer patients with DHZ.⁽³²⁾ For liver cancer, an early study revealed differences in the serum protein profiles of patients with DHZ and those with other syndromes, as well as differences before and after intervention.⁽³³⁾ Because the liver acts as the major site for protein synthesis, changes in specific proteins may aid in syndrome differentiation. Multifaceted microfeatures, such as trace elements, tumor markers, gene copy number and polymorphism, transcriptomic and proteomic profiling, and metabolomic signatures underlie the phenotypes of different syndromes in cancer patients.⁽³⁴⁾ Evidence has indicated that DHZ affects cancer phenotypes, yet the underlying molecular mechanisms of *ZHENG* are far from clarified due to the difficulties in establishing a standard *ZHENG* model conforming to CM theory for fundamental research.

Ethnopharmacological Aspect of DHZ in Cancer

Modern molecular biology technologies provide a mean to concretize the DHZ. In the era of big data, new disciplines based on high-throughput sequencing and computational models, such as network pharmacology and syndrome-based pharmacology, enable researchers to investigate the molecular basis of *ZHENG* and syndrome-guided medication from multiple dimensions.^(35,36) Compound prescriptions containing heat-clearing and dampness-eliminating (HCDE, Qing-re Hua-shi in Chinese) herbs are often used to treat cancer. Based on the weighted gene coexpression network, 12 immune-related genes were identified as potential targets of the Qingyi Huaji Formula against PC.⁽³⁷⁾ The *Coptis chinensis*-containing formula was suggested to inhibit gastric tumors by regulating glucose metabolism.⁽³⁸⁾ Huanglian Jiedu Decoction (黄连解毒汤) could inhibit eukaryotic elongation factor 2 to suppress hepatocellular carcinoma.⁽³⁹⁾ Herb monomers with HCDE effects and their derived active ingredients are widely used to screen effective anticancer agents.

For example, the commonly used rhubarb was suggested to induce lung cancer cell apoptosis, which may be attributed to emodin, resveratrol, procyanidins, etc.⁽⁴⁰⁾ *Hedyotis diffusae Herba* is frequently used in non-small-cell lung cancer (NSCLC),⁽⁴¹⁾ and is also the core herb in the Qingyi Huaji Formula against PC. *Saussureae involucreatae Herba*⁽⁴²⁾ and *Dictamnus dasycarpus* are also used to clear heat and remove dampness in cancer and their active components remain to be further clarified.⁽⁴³⁾ Many of these types of herb extracts exert antitumor activities by inducing cancer cell apoptosis, arresting the cell cycle, and inhibiting cell differentiation, proliferation, invasion, and tumor angiogenesis.⁽⁴⁴⁾

Compounds such as alkaloids, flavonoids, terpenoids, and phenolic acids possess antitumor activities through multiple signaling pathways, including the Wnt pathway, Hedgehog pathway, phosphoinositide 3-kinase (PI3K)/protein kinase B (Akt)/mammalian target of rapamycin (mTOR) pathway, mitogen-activated protein kinase (MAPK) pathway and nuclear factor-kappa B (NF- κ B) pathway.⁽⁴⁵⁾ As the most representative active ingredient of HCDE herbs, berberine has been extensively studied for its regulation of PC tumor cell metabolism, especially through targeting citrate metabolism and lactate dehydrogenase.^(46,47) Oxyepiberberine from *Rhizoma Coptidis* was found to regulate SRC, BRAF and MMP9 to inhibit NSCLC.⁽⁴⁸⁾ Matrine has a broad range of antitumor effects against many types of cancer, including cell cycle and differentiation arrest, growth and metastasis inhibition, apoptosis acceleration, angiogenesis inhibition, autophagy induction and multidrug resistance reversal.⁽⁴⁹⁾ HCDE herbs have also been proposed to act on the tumor microenvironment by inhibiting CAF proliferation and the secretion of C-X-C motif chemokines 1, 2 and 8 (CXCL1, CXCL2, and CXCL8) to suppress cancer cell motility and invasiveness as well as tumor stem cells stemness.⁽⁵⁰⁾ These findings provide evidence of the molecular mechanisms underlying DHZ from a therapeutic perspective.

Animal Model of DHZ in Cancer

Although establishing an animal model that perfectly simulates the physiological process of DHZ in cancer patients would be ideal, many obstacles must be overcome. In particular, triggering tumorigenesis after exposure to ZHENG induction factors or spontaneously developing DHZ after tumor model establishment remains challenging. However, a generally accepted mode is to combine syndrome-inducing factors with

a well-established tumor model, with the expectation that a model can be developed that simultaneously mimics the characteristics of both the disease and ZHENG. Research progress and related concerns over tumor establishment and dampness-heat induction are described as follows.

Establishment of Tumor Model

Methods for tumor model establishment are relatively mature and can be found in the literature.⁽⁵¹⁾ Here we take the PC model as a brief example: (1) a pancreatic orthotopic or subcutaneous model can be generated by inoculating human or mouse PC cell lines or tissue fragments; (2) a genetically engineered model can be developed through targeted mutation of pancreatic K-ras, Trp53, Smad4, or CDKN2A genes (single or combined mutation); (3) a chemical-induction model can be generated by dimethyl benzantracene or others;^(52,53) (4) a patient-derived organoid model can be developed, allowing individualized treatment screening and experimentation.⁽⁵⁴⁾ Among these methods, the chemical-induction model suffer from the biggest drawback due to the long time required for model establishment, along with inconsistent tumorigenesis rates and sizes, which would complicate subsequent ZHENG model establishment.

Various commercially available human cancer cell lines have been adopted to establish tumor xenograft models by utilizing immune-compromised mice, such as BALB/c nude mice and nonobese diabetic-severe combined immune deficient (NOD-SCID) gamma mice,⁽⁵⁵⁾ which have the advantages of rapid tumor growth and immune escape. However, the significant limitations of the xenograft cancer model include the lack of representative behavior of naturally occurring cancers and the absence of stromal-tumor interactions, thereby limiting the translation from bench to bedside to a certain degree.⁽⁵¹⁾ ZHENG has been suggested to be closely related to tumor immunity, and therefore, the use of immune-competent animals and respective cancer cell lines will provide further insight into the tumor microenvironment. For example, C57BL/6 mice with an integral immune system and their derived Panc02 cell line have been adopted in studies on PC immunology.^(56,57)

On the other hand, as a kind of genetic diseases, PC is found to be the most Ras-addicted cancer type, with K-Ras mutations found in most PC patients.⁽⁵⁸⁾ K-ras acts as a critical gene driving malignant progression, and induces aberrant signal pathway activation and transcription

factor production to affect cell proliferation, migration, invasion, and transition.^(59,60) The PC model established by Hingorani, et al⁽⁶¹⁾ in 2003 suggested that K-ras mutations are sufficient to initiate murine PC occurrence and progression, which resembles the clinical pathogenesis, and thus, this type of model could be applied to study the role of DHZ in PC carcinogenesis. Various cancer models have been designed for specific experimental purposes, and the use of a combination of models for exploring *ZHENG* warrants further investigation.

Induction and Establishment of DHZ

To establish an animal model of DHZ, a crucial consideration is to select appropriate model-inducing factors. *ZHENG* animal model induction is often based on the pathological changes detected by modern medical technologies, the etiology of *ZHENG* theory in CM, or a combination of both. Model development based on pathology can present clear and specific pathological changes, but the agents used for modeling are usually toxic; thus, in addition to the target organ, other unrelated physiological systems and organs may be damaged. Moreover, if researchers lose sight of the pathogenic factors based on CM theory, the pathological model can shift away from the correct interpretation of *ZHENG*.⁽⁶²⁾ The etiology-based model agrees with CM theory. However, a lack of standardized evaluation in the choice of etiological induction methods is commonly observed, leading to differences in the intensity and effectiveness of individual factor intervention and the undetermined necessity of multiple-factor intervention for successful *ZHENG* induction. This modeling method is susceptible to differences in environmental factors, intervention times, and individual traits, which can affect the degree of syndrome severity and model stability. For example, the induction agents for DHZ frequently involve the use of honey, liquor, and lard, however, due to the uneven quality of these agents, their complex composition, and inconsistent proportions, ensuring the repeatability and consistency of the model can be challenging.

In contrast to the routinely applied pathology-specific model or etiology-based model, an emerging integrated syndrome-based pathological model has been developed, which theoretically provides an etiological view closest to the understanding of diseases in CM and simultaneously takes the pathological changes into account. Integrated modeling methods have been applied to various diseases, including diabetic nephropathy, coronary heart disease,

asthma, and rheumatoid arthritis.^(63,64) Methods for inducing DHZ include fat, sweet and greasy diet intervention, herbal drugs with potent action, hot and humid environmental exposure, and biological factors.⁽⁶⁵⁾ The most common include high-fat and high-glucose diets, alcohol consumption, pathogenic bacterial infection, artificial climate chambers, and others (Table 1).⁽⁶⁶⁾ In a study on mammary tumors, single factor induction using a high-temperature and high-humidity chamber was performed to generate a dampness-heat rat model.⁽³²⁾ It is worth noting that multifactor induction with high-fat and high-glucose diets along with liquor consumption was more frequently adopted than single-agent induction. Xie, et al⁽⁶⁷⁾ reported a DHZ murine model by multifactor induction including environmental modulation induced by changes in temperature and humidity as the external factor, dietary modulation by providing a high protein diet, with lard and liquor gavage as the internal factor, and running for 30 min to simulate forced labor. It was observed that the mental state, body mass, pathology of the tongue root, serum lipopolysaccharides, TNF- α , toll-like receptor 4 signaling pathway in the colon, and aquaporin protein expression of mice were altered significantly in the DHZ group compared with the control group. Wang, et al⁽⁶⁸⁾ reported two similar models induced by a shared combination of a high protein diet, syrup, liquor, and an artificial climate chamber, in addition, arachis oil and folium sennae were respectively added into the intervention of two independent groups. Gut microbiota structure between the 2 groups were compared, results showed that the DHZ induced by the two methods presented different gut microbiota compositions.

It can be seen that although the same *ZHENG* can be developed through various methods, different combinations of induction factors will stimulate different microreactions in the body. The discrepancy under differential factor-induction may be explained by the different internal organs that are damaged. According to viscera attributes (Zang-fu in Chinese), the dampness-heat niche is attributed to Pi, Wei (Stomach), Dachang (Large Intestine), Gan and Dan (Gallbladder), Pangguang (Bladder), and others.⁽⁶⁹⁾ The modeling methods for the aforementioned subtypes of DHZ are still inconclusive; however, it is clear that a *ZHENG* model, established by a unified method is reasonable for comparative evaluation and further in-depth research.

In addition, these methods mostly originate from the etiological understanding of human disease according to CM theory. However, although the commonly used animal

Table 1. Methods for Developing Various Animal Models with Dampness-Heat Syndrome

Syndrome induction methods	Animals, M/F	References
Cholesterol, hot pepper powder, fat	Rabbit, M/F	Zhang, et al, 2002 ⁽⁷³⁾
Honey, lard, climate chamber, <i>S. typhimurium</i>	SD rat, M/F	Guo, et al, 2003 ⁽⁷⁴⁾
High-fat high-sugar feed, climate chamber, enterotoxigenic <i>E. coli</i>	SD rat, M/F	Mu, et al, 2003 ⁽⁷⁵⁾
High-fat high-sugar feed, Dahuang Decoction, climate chamber, <i>S. typhimurium</i>	SD rat, M/F	Yan, et al, 2005 ⁽⁷⁶⁾
Honey, lard, climate chamber, Fuzi Ganjiang Decoction, <i>S. typhimurium</i>	SD rat, M/F	Liao, et al, 2005 ⁽⁷⁷⁾
Honey, lard, climate chamber, liquor or Fuzi Ganjiang Decoction, <i>E. coli</i>	SD rat, M/F	Tan, et al, 2009 ⁽⁷⁸⁾
Honey, lard, liquor	C57 mouse, M	Dai, et al, 2010 ⁽¹⁰⁾
Honey, lipa, liquor, climate chamber, sodium salicylate	SD rat, M/F	Wang, et al, 2015 ⁽⁷⁹⁾
Glucose, lard, climate chamber	SD rat, M/F	Wu, et al, 2015 ⁽⁸⁰⁾
High protein feed, sucrose water, climate chamber, arachis oil, folium sennae, liquor, lipopolysaccharide	BALB/c mouse	Wang, et al, 2017 ⁽⁶⁸⁾
Climate chamber	SD rat, F	Dong, et al, 2018 ⁽³²⁾
High protein feed, lard, liquor, climate chamber, forced labor	BALB/c mouse, M	Xie, et al, 2018 ⁽⁶⁷⁾
Sucrose water, hot pepper, dry ginger oil, liquor, climate chamber, enteroinvasive <i>E. coli</i>	SD rat, M/F	Wang, et al, 2019 ⁽⁸¹⁾
Cholesterol, hot pepper powder, fat	Rabbit, M/F	Zhang, et al, 2002 ⁽⁷³⁾
Honey, lard, climate chamber	Wistar rat, M	Guo et al, 1988 ⁽⁷⁰⁾
Honey, lard, climate chamber, <i>E. coli</i> endotoxin	New Zealand white rabbit, M/F	Wang et al, 1990 ⁽⁷¹⁾
High-fat high-sugar feed, climate chamber, <i>S. typhimurium</i> or <i>E. coli</i>	Wistar rat, M/F	Wu et al, 1999 ⁽⁷²⁾

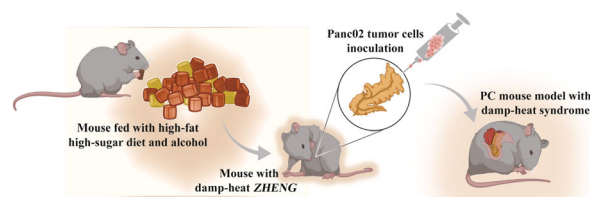
Notes: M/F: male/female; SD: Sprague-Dawley; *E. coli*: *Escherichia coli*; *S. typhimurium*: *Salmonella typhimurium*

models share a relatively high genetic and phenotypic homology with humans, it is unclear whether the etiology of ZHENG in animals is similar to that in humans, and there is also a lack of appropriate evaluation methods.

Previously, we constructed a DHZ PC model by subcutaneously inoculating the Panc02 cell line into C57BL/6 mice fed honey water, lard, and liquor.^(10,82) After model establishment, data on feces, hair, food and water consumption behavior, psychomotor ability, and body temperature were recorded for syndrome severity assessment. Our report described the establishment of dampness-heat PC model with sticky and loose feces, liparotrichia, a reduction in food intake, mental

retardation, and slightly higher skin temperature than the standard control. These changes in behavior were consistent with the clinical characteristics of DHZ in CM theory. To strengthen the credibility and recognition of the DHZ model, appropriate evaluation criteria must be established to objectively assess and quantify the subjective indicators of CM syndrome and to facilitate the standardization of the ZHENG models.

Based on the model we established, the relationships between PC and CM syndrome were investigated by establishing "tumor-first" and "syndrome-first" models to analyze the impact of the order of ZHENG occurrence on tumor development. Our results showed that DHZ occurring before tumor transplantation could promote tumor growth, while DHZ occurring after tumor transplantation had no significant effect on tumor growth.⁽²⁷⁾ At the same time, analysis of tumor-associated macrophages and inflammatory factors suggested that there was an intimate relationship between DHZ and the tumor microenvironment. Our persistent works have established a prototype of the DHZ model of PC (Figure 1) and provide preliminary experience for subsequent studies.

**Figure 1. Diagrammatical Sketch for Establishing Dampness-Heat ZHENG Mouse Model of Pancreatic Cancer**

As reported in other study, tumor locations, clinical stages, and treatment strategies can affect the ZHENG characteristics in cancer patients,⁽⁸³⁾ illustrating the complex interactions between tumors and ZHENG. Currently, there have been relatively few studies focusing on ZHENG in cancer, especially fundamental studies, and the mechanisms of ZHENG evolution and their relationships with modern pathophysiology have not been fully elucidated. Clinical ZHENG differentiation in cancer patients is mainly dependent on the clinician's subjective judgment, as objective criteria are lacking, which hinders a systematic study of ZHENG in cancer.⁽⁸⁴⁾ According to ZHENG elements, classification modeling is worthy of study to promote ZHENG differentiation and standardization in research.⁽⁸⁵⁾ In short, cancer is a complex disease caused by multiple factors, and ZHENG

is a highly generalized pathological process during a particular stage of the disease. Therefore, the complexity of the integrated cancer-syndrome model poses challenges to its establishment and evaluation.

Measures for Building Dampness-Heat Model Evaluation System

Finally, the ZHENG model needs to be evaluated and tested using uniform standards to ensure its reproducibility and credibility for subsequent research. The existing ZHENG models and evaluation methods are relatively outdated in terms of standardization and objectivity, making it difficult for researchers to use these models and limiting their recognizability. Model evaluation measures that combine the macroscopic ZHENG differentiation in CM with microscopic biological and pathological evidence can improve the objectivity and persuasiveness of this research.⁽⁸⁶⁾

The evaluation of DHZ, from a macroscopic perspective, includes observing animal behaviors, including their mental state, hair, diet, skin and urine color, fecal texture, weight, body temperature, and tongue. Grading quantification and standardized syndrome scales applied in cancer patients or nontumor diseases can be used for reference in evaluating the DHZ cancer animal model.^(4,87,88) The early proposed ZHENG identification criteria, especially for rats and mice, are of great value for model evaluation,⁽⁸⁹⁾ and quantifying the diagnostic performance of these features through recognized indicators will reduce subjective bias and increase credibility. Automated feature extraction based on machine learning is an objective and fast technique for ZHENG classification⁽⁹⁰⁾ and is worthy of further development for animal ZHENG model evaluation in the future.

From the microscopic aspect, it is possible to adopt tissue sectioning and molecular techniques to detect morphological and functional changes in organs, such as the pancreas, liver, intestine, and even the tongue. Many molecular and pathological changes related to clinical symptoms and ZHENG can be integrated into the ZHENG model assessment.⁽⁹¹⁾ The above mentioned molecular landscape of DHZ mostly reflects the aspect of the immunity signature. A high-fat, high-sugar diet and alcohol intake can lead to metabolic changes that mimic abnormal metabolism under dampness-heat conditions. Therefore, the detection of metabolite profiles related to metabolic disturbances may contribute to model

evaluation. Specific indicators need to be screened in future work and validated through experiments. From a clinical perspective, the correlation analysis between DHZ and routine clinical, biochemical indicators and pathological manifestations can be studied to identify the typical microscopic indicators, such as the systemic inflammatory response index,⁽⁹²⁾ thereby improving the accuracy of evaluation and strengthening the clinical value of basic research. Together, the findings in the literature indicate that the systems biology concept of modern medicine coincides with the holistic view of CM,⁽⁹³⁾ and the utilization of multiomics strategies, including genomic, transcriptomic, proteomic, epigenomic, and metabolomic analyses, can provide a powerful resource for clarifying the essence of ZHENG and evaluating ZHENG models.

Gaps and Bridges

CM and contemporary Western medicines vary substantially in their understanding of disease etiology and pathogenesis. Contemporary Western medicine emphasizes evidence-based studies, and the pathogenesis is measured by specific biochemical indices or imaging examinations. CM, on the other hand, emphasizes a holistic view and ZHENG differentiation. Disease etiology in CM theory includes an overall consideration of the condition using empirical medicine, including diet, emotions, exogenous pathogenetic factors, internal factors, etc. The CM view of disease development often relates to an imbalance of "the eight principal syndromes of differentiation," including yin and yang, external factors and internal factors, cold and heat, as well as hypofunction and hyperfunction. Additionally, other ZHENG differentiation methods used in CM, include the Zang-Fu organ functions theory, the four body strata theory, including defense (Wei), vital energy (qi), nutrient (Ying), and blood (Xue) transformation, and the six meridians tropism. The abstract nature of the ZHENG terminology makes it challenging to concretize and reproduce a ZHENG model. Furthermore, the ambiguous terminology and uncertainties regarding its biological basis challenge the ability to use evidence-based evaluations to delineate CM theory. Considering the inconsistent recognition and divergence in understanding the particularities of ZHENG among different scholars, the factors needed to develop ZHENG models may vary across diseases with identical ZHENG manifestations.

It is worth noting that the emerging comprehensive multifactor model combining both disease and ZHENG

features provides a better simulation of CM *ZHENG* characteristics in disease⁽⁶²⁾ and is becoming a highly recognized modeling approach. However, the standardization and specification of this multifactor model need to be further explored and verified. A high-fat diet is considered an important cause of DHZ in CM theory and is correlated with metabolic syndrome, which is a risk factor for cancer.^(94,95) The latest research has suggested that a high-fat diet induces competitive lipid metabolism between tumor cells and immune cells and can remodel the tumor microenvironment. The ensuing T cell infiltration and dysfunction allow tumor cells to escape immune surveillance.⁽⁹⁶⁾ The multifactor comprehensive model of cancer induced by a high-fat, high-glucose diet and liquor consumption is not only in line with CM theory but also consistent with evidence-based medicine, suggesting the feasibility and suitability of this method for the establishment of a DHZ model. To ensure reproducibility, the use of commercially manufactured diets should be used for research purposes, as they are readily available and provide the advantages of being germ-free, gnotobiotic, specific-pathogen-free, and all ingredients are verified and disclosed. For liquor interventions, the adoption of a fixed concentration of edible ethanol eliminates the possible interference of multiple components in liquor and would facilitate widespread accessibility of the model. Different types of lipids, such as butter, lard, and soybean oil, and various sugar types, such as glucose, fructose, and galactose, lead to differences in glucose and lipid metabolism,^(97,98) which should also be taken into consideration during model establishment. Furthermore, the differential behavioral and physiological responses to a high-fat diet and preference for alcohol intake between male and female animals⁽⁹⁹⁾ should be carefully considered during animal selection.

Since the specific meanings of abstract concepts in CM have not been clarified, the causal relationship between diseases and *ZHENG* has not been resolved. Moreover, the selected *ZHENG* indicators and evaluation methods are often affected by the interest in a specific subject, so there is almost no consensus that can be used for the establishment of *ZHENG* models, further limiting the comprehensive reproducibility of an animal model for basic science research.⁽¹⁰⁰⁾ However, as mentioned above, by identifying the similarities between CM and contemporary Western medicine, based on current research evidence, a *ZHENG* animal model can begin to be developed that addresses both the etiology and pathogenesis of cancer. With additional research,

the *ZHENG* model is expected to be gradually optimized, and vice versa. The standard *ZHENG* model and evaluation system will ensure the scientific validity and reproducibility of research and pave the way for revealing the mechanism of cancer and developing new clinical therapies.

Conclusion

Taking advantage of *ZHENG* differentiation in CM and strengthening fundamental research will help to supplement clinical solutions to treat cancer. The proposal of DHZ as the core pathogenesis and the formulation of relevant therapeutic strategies are crucial achievements in applying CM to PC treatment. Further exploration of the essence of *ZHENG* will be the key to a breakthrough, and the establishment of a *ZHENG* model will lay the foundation for extensive research. Here we attempted to conceive a relatively standard DHZ model of cancer and a systematic methodology for analyzing the theoretical basis and modern connotation of *ZHENG* differentiation in CM. The unified *ZHENG* model and evaluation system are also critical to ensure research quality, promote academic exchange and realize better precision medicine strategies. This paper introduces the current understanding of DHZ in pancreatic treatment cancer, discusses progress on the establishment of a DHZ model, and proposes some practicable model standardization measures. We aim to collate the limited information for public reference and hope to work together on *ZHENG* research in cancer in the future.

Conflict of Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Author Contributions

Chen Z (<http://orcid.org/0000-0002-4502-0801>) conceptualized and organized this review, Jiao JY (<https://orcid.org/0000-0003-4435-9931>) analyzed literature and drafted the manuscript, Chen CS revised the manuscript, Cao ZQ and Chen LY collected literature. All authors read and approved the final manuscript.

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